

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: looking at a RCBO

Assistant: To help you select the right polymer material for your RCBO application, could you please clarify:

1. What specific part of the RCBO are you looking to use the polymer for? For example, is it the housing, internal switch mechanism, terminal block, or another component?
2. Where will this RCBO be used in your solution? Is it for residential, commercial, or industrial electrical panels, or another end-use scenario?

For instance, RCBO housings are often used in consumer units for homes, while switch actuators may be used in industrial control panels. Your answers will help me identify the most suitable material options.

User: housing,residential and indoor

Assistant: Thank you for clarifying. To further narrow down the material options for the RCBO housing:

1. What is the minimum wall thickness of your housing part?
2. Do you have any preferred base materials (such as specific polymers) that you would like to use for this application?
3. Are you open to alternative material suggestions if your preferred material is not available or optimal?

These details will help ensure the material meets your requirements for electrical insulation, mechanical strength, and processing.

User: 0.8mm and open to suggestions

Assistant: Here are the key material requirements (CTQs) assumed for your RCBO housing application:

1. Minimum wall thickness: 0.8 mm (as specified)
2. Flammability classification: UL94 V-0 at 0.8 mm
3. Glow Wire Flammability Index (GWFI): 960°C at 0.8 mm
4. Comparative Tracking Index (CTI): 600 V
5. Processing requirement: Injection molding
6. Environmental exposure: Halogen-free, flame retardant
7. Filler: Glass fiber reinforcement

Are all of these requirements correct for your application? Please confirm if these match your needs for the RCBO housing.

User: also need glass fib 25 to 35

Assistant: Thank you for confirming the requirements and specifying the glass fiber content (25% to 35%).

To ensure the material fully meets your needs, could you please provide or confirm the following additional requirements:

1. Relative Thermal Index - Electrical (RTI Elec): What is your target value for long-term electrical thermal rating?

2. Relative Thermal Index - Mechanical Strength (RTI Str): Is there a specific value you require for mechanical strength over time?
3. Glow Wire Ignition Temperature (GWIT): Is there a minimum GWIT value you need for the housing?

These properties are important for electrical safety and long-term performance. Please confirm or provide your target values for these, or let me know if you are unsure.

User: i dont have this information you can recommend gades based on above ctqs

Assistant: Based on your requirements for RCBO housing (residential, indoor, minimum wall thickness 0.8 mm, UL94 V-0, GWFI 960°C, CTI 600 V, halogen-free flame retardancy, glass fiber 25-35%, injection molding), here are potential grade options that may be suitable:

1. FRIANYL® A3 H GF25 V0 NC 1102/EA – PA66, 25% glass fiber, natural color, meets all critical safety and electrical CTQs at 0.8 mm.
2. Zytel® FR95G25V0NH NC010 – PA66, 25% glass fiber, natural color, higher thermal indices for robust performance.
3. Zytel® HTNFR42G30NH NC010 – PPA, 30% glass fiber, natural color, exact UL approvals at 0.8 mm, strong thermal/chemical stability.
4. FRIANYL® A3 GF30 V0 GY 7035 – PA66, 30% glass fiber, grey color, meets all CTQs at 0.8 mm.
5. Zytel® HTNFR52G35NHF BK337 – PPA, 35% glass fiber, black color, meets all CTQs at 0.8 mm.

These grades are selected based on technical data and may be potential options for your application. Please ensure compliance with all applicable standards and

regulations. Final material selection and product design decisions should be made based on your specific requirements and testing.

*Note: This recommendation is generated by an AI assistant and should be considered purely informational.*